Bond Ordering in Flux Phases

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Based on arXiv:2402.09577

Introduction

Periodic arrangement of magnetic fluxes can lead to new phases and phenomena.

- Single particle effects: spin orbital coupling(SOC)$\rightarrow$ Quantum Anomalous Hall (QAH) effect, Quantum Spin Hall Effect (QSH)...
- Many body effects: Stagger flux phases, d-density wave state (DDW), Loop Current phase (LC), QAH, QSH...
- Possible novel ground state in flux phases?
  The nature of quantum phase transitions?

Two-site Example

- Consider a two-site model with spinless electrons :

$$H = -t(c_1^\dagger c_2 + c_2^\dagger c_1) + V n_1 n_2$$

- Mean Hamiltonian by condensing $\hat{J}_{12} = \imath(c_1^\dagger c_2 - c_2^\dagger c_1)$, with $J_{12} = \langle \hat{J}_{12} \rangle$

$$H_{eff} = -(t + \imath V J_{12}) c_1^\dagger c_2 + h.c + ... = -\frac{t}{\cos(a_{12})} e^{\imath a_{12}} c_1^\dagger c_2 + h.c. + ...$$

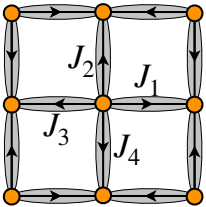
- The phase a_{ij} on the link can be eliminated by a gauge transformation.
  For lattice model with net flux within a plaquette, the flux cannot be gauged away.
  Flux pattern $\xrightarrow{Gauge\ invariance}$ fermionic ground state.
- Our work: the flux pattern does not uniquely determine the fermionic ground state. Phase is not redundant as it renormalizes hopping amplitude.

One-band Model

$$\hat{H} = -t\sum_{\langle ij\rangle}(\hat{c}_i^\dagger \hat{c}_j + h.c.) - t'\sum_{\langle\langle ij\rangle\rangle}(\hat{c}_i^\dagger \hat{c}_j + h.c.) + V\sum_{\langle ij\rangle}\hat{n}_i\hat{n}_j$$

We perform mean-field studies of the Hamiltonian by condensing the bond-current operators.
Central question: Phases where fluxes are translationally invariant but bond orders are not?

- Consider orders within the $\sqrt{2}\times\sqrt{2}$ unit cell(4 links).



- The phase diagram can be determined by minimizing the free energy wrt J's for given filling factor ν and interaction V.

Mean Field Phase Diagram

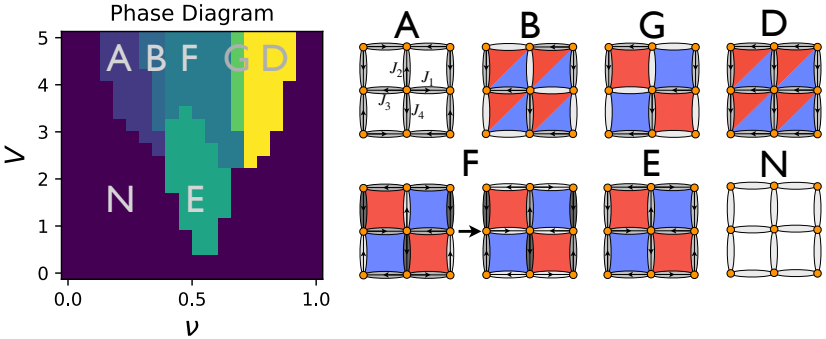
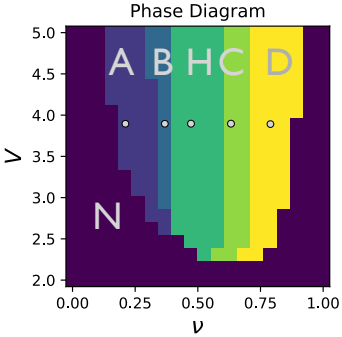
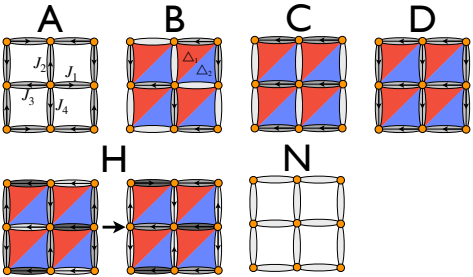


Figure: Mean Field Phase Diagram in the (V, ν) plane.

- Fermi liquid states without flux: phase A and phase N
- Θ_{II}-like phases: phase B and phase D
- Staggered flux phases: phase E, F and G

Mean Field Phase Diagram (Zero Flux Subspace)

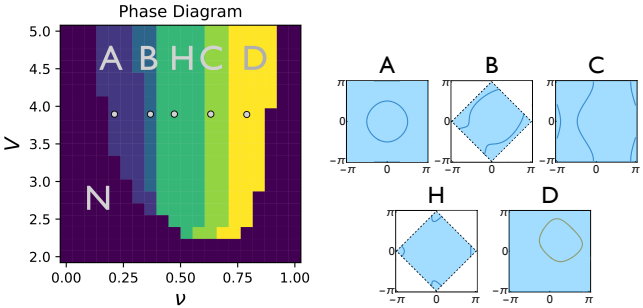




Focus on the subspace of flux phases with vanishing net flux per unit cell, as the flux patterns respect translational symmetry.

- Near the half filling, the staggered flux phases → phase C and H.
- Phase H is translationally-broken; phase C are translationally-invariant.

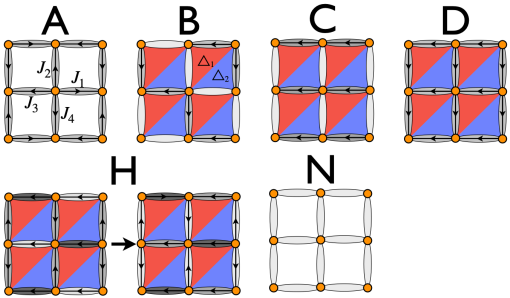
Fermi surfaces



- A: Circular Fermi surface.
- B&C: Strip-like ordering.
- H: Electron pockets at $(\pi, 0)$ and $(0, \pi)$
- D: Non-circular Fermi surface.

Summary and Conclusions

- We have studied a spinless-fermion model on a square lattice that gives rise to various flux phases that breaks time-reversal symmetry.
- In this one-band model, flux pattern cannot uniquely determine the fermionic ground state.



Thanks for your attention!